

Solution Architecture

Date	Team ID	Project Name	Maximum Marks
		SB Stocks	4 Marks

Detailed description of the system architecture.

Architecture Overview:

The SB Stocks application follows a classic three-tier web architecture:

- 1. Presentation Layer (Frontend) - HTML/CSS/JS with Bootstrap
- 2. Application L

Component descriptions:

Frontend Components:

- Login/Register Page: User authentication forms
- Dashboard: Ma

Backend Components:

- Auth Module: Registration, login, session management (Flask-Login)
- Stock API Servi

External Integrations:

- Alpha Vantage API: Real-time and historical stock market data

Data Flow:

User -> Browser (HTML/CSS/JS) -> Flask Server (Routes + Business Logic) -> Alpha Vantage API (Stock Data) + SQLite DB (User Data, Portfolios, Transactions)

Security Architecture:

- Password hashing with bcrypt
- Session-based